Mosammat Tabasum Mastaba Isma

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SUMMARY

Computer and Communication Engineering graduate (3.758 CGPA) with hands-on computer vision and deep learning research, building end-to-end image segmentation pipelines in Python and PyTorch, from dataset cleaning to GPU-accelerated training and evaluation (90.46% mIoU). First author of a paper on multimodal crack segmentation and geometric quantification accepted for oral presentation at IEEE ICISET 2026. Teaching Assistant mentoring 40+ students; coordinated events for 200+ participants as CCE Club Female Chapter GS. Keen to apply imagery-based AI to agriculture and field operations.

PUBLICATIONS

- **BAGFD-Net: Multimodal Boundary-Aware Fusion for Crack Segmentation and Geometric Quantification.** First author. Accepted for oral presentation, 5th International Conference on Innovations in Science, Engineering and Technology (IEEE ICISET 2026), 2026.

EXPERIENCE

Undergraduate Thesis | IIUC **2026**

- Built an end-to-end PyTorch pipeline for road crack segmentation (dataset cleaning, Swin-Tiny + attention-gated decoder, GPU training on Kaggle/Colab, evaluation), reaching 90.46% mIoU; accepted for oral presentation at IEEE ICISET 2026.

Teaching Assistant | Data Structures & Algorithms | IIUC **2025 – 2026**

- **Mentored 40+ students** on complex algorithmic logic and C++ programming efficiency.
- **Debugged and reviewed code** for undergraduate labs, ensuring adherence to clean code and OOP principles.

Teaching Assistant | Java | IIUC **2024**

- **Provided technical guidance** to students, resulting in a measurable improvement in class project completion rates.
- **Debugged student code and evaluated logical efficiency**, building hands-on experience directly transferable to model performance evaluation.

Chief Executive (GS) | CCE Club Female Chapter | IIUC **2022 – 2023**

- **Coordinated with senior faculty and industry professionals** to manage technical events for 200+ participants.

PROJECTS

- **Crack Segmentation (BAGFD-Net / Thesis):** Multimodal crack segmentation with geometric quantification; 90.46% mIoU.
- **Blockchain-Integrated Tutor App:** Designed a platform that uses blockchain to make tutor-student interactions secure and transparent.
- **MIR Online Shopping:** Built a full-stack e-commerce website with interactive user interfaces and shopping logic.
- **Campus Voice App:** Developed an Android app for communication and feedback within the university campus.
- **Laser Guard (IoT):** Built an Arduino-based IoT security system using laser detection.

EDUCATION

BSc in Computer and Communication Engineering (CCE) | IIUC **2022 – 2026**

Thesis Completed (GPA – 4.00) | 8th Semester GPA – 3.956 | CGPA – 3.758

Presidency International School (British Council, Cambridge) **2015 – 2020**

- GCE A Levels (2020) – Science
- GCE O Levels (2018) – Science

TECHNICAL SKILLS

Languages: Python (Proficient), C/C++, Java

Web Development: HTML, CSS, JavaScript

Core Fundamentals: Object-Oriented Programming (OOP), Data Structures & Algorithms

AI & Data Science: Machine Learning, Deep Learning (PyTorch, Swin-Transformer, CNNs), Computer Vision (image segmentation, attention mechanisms), Model Evaluation (mIoU), Dataset Cleaning, GPU-accelerated Training (Kaggle, Google Colab), NLP Fundamentals

CERTIFICATES

- **3rd Position**, IIUC TECHFEST 2025 — Mobile Games and Apps Development Competition.
- **Best Rookie Team Position**, Inter Semester Programming Contest 2022, organized by CCE Club, IIUC.
- **Daily Star Awardee** on O Levels with highest achieving grades.

REFERENCES

Zia Uddin Ahmed
Senior Software Engineer, Panacea Plt
Kuala Lumpur, Malaysia

Md Humayun Kabir
Technical Trainer, New Vision IT